

Question 1 (Data Selection for Supervised Learning)

In this question, each data point has two features. Each plot represents a specific data distribution and the decision boundary of the **linear classifier** being trained to perform binary classification between **circles** and **crosses**.

In all the subsequent questions, assume the data-selection algorithm selects **exactly 4 examples**.

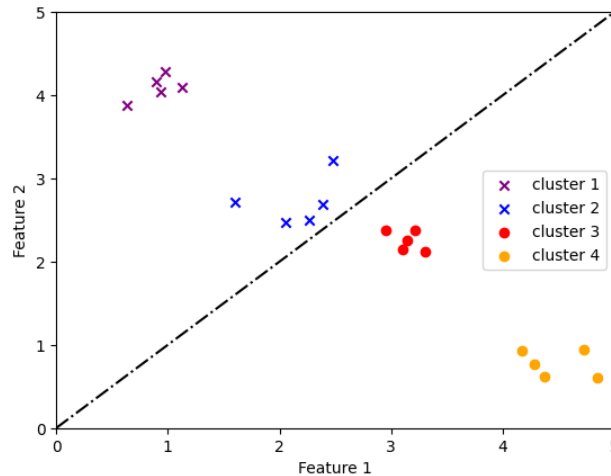


Figure 1: Data Selection (part 1 and 2)

1. **(5 points)** Refer to Fig. 1. Which clusters do the data points selected by **Forgetting Score Algorithm** belong to?

- A. Cluster 1
- B. Cluster 2
- C. Cluster 3
- D. Cluster 4
- E. Cluster 1, Cluster 2
- F. Cluster 1, Cluster 3
- G. Cluster 1, Cluster 4
- ☒ H. Cluster 2, Cluster 3
- I. Cluster 2, Cluster 4
- J. Cluster 3, Cluster 4
- K. Cluster 1, Cluster 2, Cluster 3
- L. Cluster 1, Cluster 2, Cluster 4
- M. Cluster 1, Cluster 3, Cluster 4
- N. Cluster 2, Cluster 3, Cluster 4
- O. Cluster 1, Cluster 2, Cluster 3, Cluster 4

*forgetting = close to boundary,
more difficult examples*

2. (5 points) Refer to Fig. 1. Which clusters do the data points selected by **CRAIG** belong to?

- A. Cluster 1
- B. Cluster 2
- C. Cluster 3
- D. Cluster 4
- E. Cluster 1, Cluster 2
- F. Cluster 1, Cluster 3
- G. Cluster 1, Cluster 4
- H. Cluster 2, Cluster 3
- I. Cluster 2, Cluster 4
- J. Cluster 3, Cluster 4
- K. Cluster 1, Cluster 2, Cluster 3
- L. Cluster 1, Cluster 2, Cluster 4
- M. Cluster 1, Cluster 3, Cluster 4
- N. Cluster 2, Cluster 3, Cluster 4
- ☒ O. Cluster 1, Cluster 2, Cluster 3, Cluster 4

select weighted subset that closely
estimates full gradient

↳ find exemplars

↳ weigh by size of cluster

each cluster about the same size

↳ select all

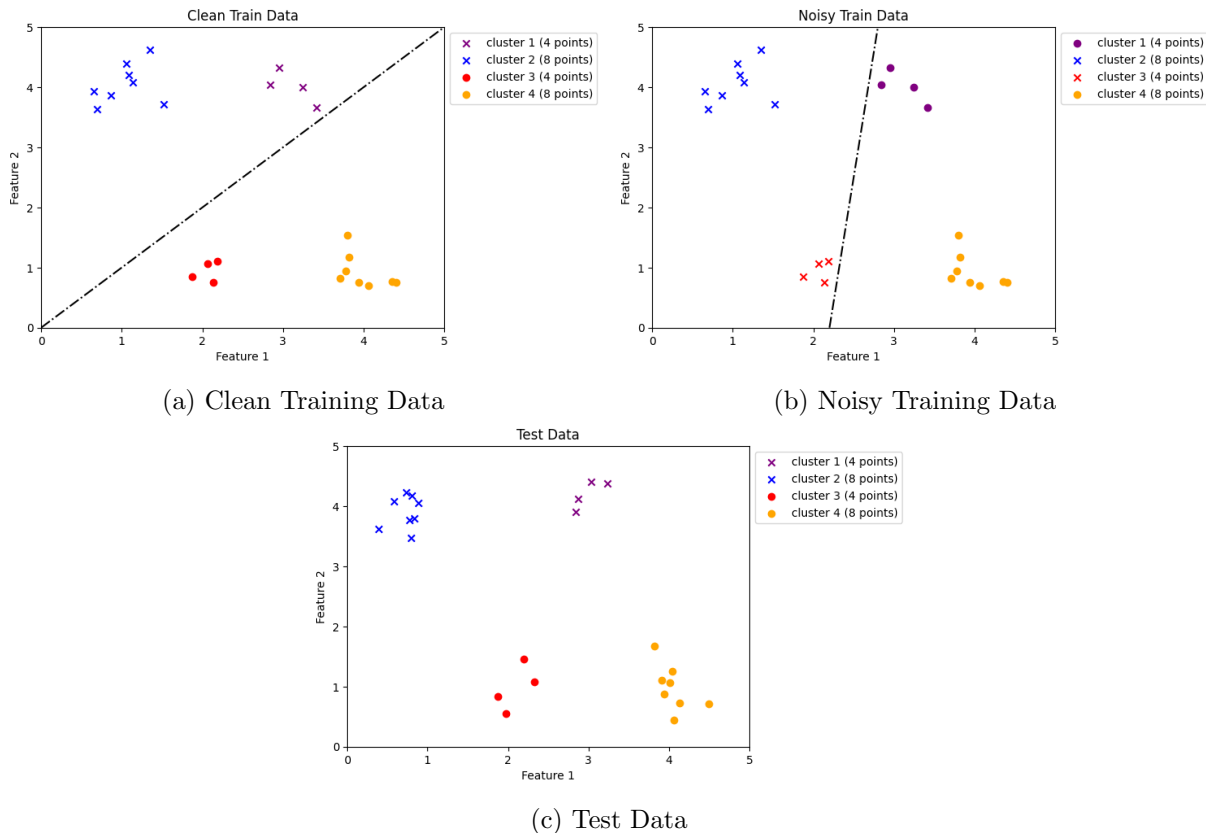


Figure 2: Data Selection in the presence of Noisy Labels (part 3 and 4)

For the following questions, refer to Fig. 2. Here Fig. 2a shows the clean training data and Fig. 2b shows the noisy training data available to the data-selection algorithms in this section. The dotted line indicates the decision boundary of a model. In the noisy training data, the points in Cluster 1 and Cluster 3 have their labels flipped as shown in Fig. 2b. Fig. 2c shows the test data.

Assume that the models trained on the selected subsets and the full noisy training data achieve 100% training accuracy.

3. (5 points) Is the subset selected by the **Forgetting Score Algorithm** likely to achieve the same test accuracy as training on the **full noisy training data**?

Note: For simplicity, assume that the decision boundary shown in 2b is used for data subset selection with the Forgetting Score Algorithm.

A. Yes

☒ B. No

better at difficult examples

4. (5 points) Is the subset selected by the **CRAIG** likely to achieve similar test accuracy to training on the **full noisy training data**?

☒ A. Yes

B. No

quicker than training full data, but same accuracy

Question 2 (Data Selection for Contrastive Learning)

Given the following data and latent classes, select a subset of size 4 using SAS.

Compare similarity between features using negative L1-Norm for simplicity. Thus, the optimal subset S_* is defined here as:

$$S_* = \operatorname{argmin}_S \sum_{k \in K} \sum_{i \in S_k} \sum_{j \in V_k \setminus S_k} \|x_i - x_j\|_1$$

- where k refers to the latent class of the examples
- V_k refers to the set of examples in latent class k in the full data
- S_k refers to the set of examples in latent class k in the selected subset
- $S = \cup_{k \in K} S_k$

Find S_* i.e. the subset of size 4 that minimizes the above objective function.

- $K = \{0, 1\}$
- $V_0 = \{2, 3, 4, 5, 7, 9\}$
- $V_1 = \{1, 6, 8, 10, 11, 12\}$

The examples are as follows:

$x_1 = [0.0, 0.0, 1.5, 3.0, -0.05]$, its latent class = 1
 $x_2 = [4.0, 6.0, 0.0, 0.0, 0.05]$, its latent class = 0
 $x_3 = [6.0, 4.0, 0.0, 0.0, -0.05]$, its latent class = 0
 $x_4 = [6.0, 6.0, 0.0, 0.0, -0.05]$, its latent class = 0
 $x_5 = [8.0, 6.0, 0.0, 0.0, -0.05]$, its latent class = 0
 $x_6 = [0.0, 0.0, 1.5, 4.5, -0.05]$, its latent class = 1
 $x_7 = [6.0, 6.0, 0.0, 0.0, 0.05]$, its latent class = 0
 $x_8 = [0.0, 0.0, 1.5, 6.0, 0.05]$, its latent class = 1
 $x_9 = [6.0, 8.0, 0.0, 0.0, 0.05]$, its latent class = 0
 $x_{10} = [0.0, 0.0, 1.0, 4.5, 0.05]$, its latent class = 1
 $x_{11} = [0.0, 0.0, 2.0, 4.5, -0.05]$, its latent class = 1
 $x_{12} = [0.0, 0.0, 1.5, 4.5, 0.05]$, its latent class = 1

Select indices of all examples that will be selected.

4, 6, 7, 12 have code does it for me on github

Question 3 (Spurious Correlations)

Introducing SpuCoMNIST: In this homework question, we delve into SpuCoMNIST, a dataset designed for colored digit recognition. SpuCoMNIST is an extension of the classic MNIST dataset, where additional colored backgrounds have been introduced as spurious features. This task involves recognizing digits while dealing with these added complexities. With SpuCoMNIST, one can explore various factors, such as the number of classes, group sizes, and the ability to control the magnitude and variance of the spurious features. Furthermore, it offers the opportunity to experiment with label noise and noise affecting the core features.

Here we initialize SpuCoMNIST with 5 classes i.e (0,1); (2,3); (4,5); (6,7); (8,9) as specified here: (<https://colab.research.google.com/drive/1FfTwWQScrE4skdGyKj6v4IFN3GJAkcjA?usp=sharing>) and investigate the effect of different parameters controlling spurious correlation on the accuracies of various groups.

1. **(5 points)** Is the accuracy of examples in group (1, 1) higher or lower or equal to that of examples in group (0, 1)?

- ☒ A. Higher
- B. Lower
- C. Equal

2. **(5 points)** (Continuing from Part 1) Why?

- A. Group (1,1) is a minority in class 1; Group (0,1) is the majority in class 0.
- ☒ B. Group (1,1) is the majority in class 1; Group (0,1) is a minority in class 0.
- C. Group (0,1) is mislabeled.

3. **(5 points)** Now vary the spurious correlation strength from (1) 0.995 to (2) 0.975 to (3) 0.9. (First change from (1) to (2) then (2) to (3) to answer this question). What is the effect of decreasing the spurious correlation strength on the accuracy of group (0,1)?

- A. Increases then decreases
- B. Decreases then increases
- C. Increases then stays the same
- D. Decreases then stays the same
- ☒ E. Increases
- F. Decreases
- G. Stays the same

0.995 $\rightarrow$ 0.0
0.975 $\rightarrow$ 0.9956
0.9 $\rightarrow$ 96.9267

4. **(5 points)** Set spurious correlation strength to 0.995 again. Now vary the spurious feature difficulty from (1) **MAGNITUDE SMALL** to (2) **MAGNITUDE MEDIUM** to (3) **MAGNITUDE LARGE**. (First change from (1) to (2) then (2) to (3) to answer this question). What is the effect of this on the worst-group accuracy?

- A. Increases then decreases
- B. Decreases then increases
- C. Increases then stays the same
- D. Decreases then stays the same
- E. Increases
- ☒ F. Decreases
- G. Stays the same

1) 94.933 (2.0)
2) 14.6095 (4.1)
3) 0.0 (6.1)

5. **(20 points)** Set spurious correlation strength to 0.995 and spurious feature difficulty to **MAGNITUDE LARGE**. Implement JTT's Group Inference and Robust Training. Add the code for this implementation in the provided blocks in the notebook linked above. Run this and save the notebook with both code and outputs as a PDF. Tune the hyper-parameters of JTT's group inference and/or robust training phases as needed to achieve an improvement in worst-group accuracy. Report the worst-group accuracy you achieved after using JTT. Print the entire notebook with your code and output as a pdf and submit this PDF.

Question 4 (Poisoning Attack)

In a targeted poisoning attack, the goal is to misclassify the target x_t as the adversarial class y_{adv} . Specifically, the poisons in the training data need to pull the representation of the target toward the poison class.

Consider a linear regression model $y = aX + b$, with parameters $a = [0.3, 0.1, 0.2]$ and $b = 0$. Suppose we use a threshold of 1 for predicting labels. If the model output value is larger or equal to 1, the data is classified as 1 otherwise it is classified as 0. For simplicity, use MSE loss to update the model. The attacker has access to three of the data points in $\mathcal{V}$: $x_1 = [1, 1, 3]$, $y_1 = 1$; $x_2 = [1, 2, 2]$, $y_2 = 0$; $x_3 = [2, 1, 0]$, $y_3 = 1$. Suppose the target (test) data point $x_4 = [3, 0, 5]$, $y_4 = 1$. The goal is to misclassify y_4 as 0. To calculate dissimilarity between data points x and y , use l2 norm $\|x - y\|_2 = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$ for both questions below.

1. Suppose you want to conduct a targeted clean-label data poisoning attack on the model, and the perturbation you intend to do is $\delta = 1$ on the first element of the x vector. Assume you can perturb the data only once. If you can only select one data point to perturb, which data point would you select? What would the corresponding y label be? *choose x_1*
2. Suppose that you want to optimally modify $V_p = \{x_3\}$ to achieve your goal in a clean label attack, and norm of the perturbation is bounded by $\varepsilon = 0.5$, i.e., $C = \{\delta \in \mathbb{R}^{n \times m} : \|\delta\|_\infty \leq \varepsilon, \delta_i = 0, \forall i \neq V_p\}$. What's the resulting value for x_3 ? (Note that in the real world, the class of poisoned examples' are typically aligned with the target class.) *$x_3 = [2.5, 0.5, 0.5]$*
3. (4 points) In the EPIC algorithm, the model will iteratively drop the data points in small clusters in the gradient space. What's the correct reasoning for this?

- A. Data points with isolated gradients are unforgettable and can be safely removed from the dataset
- B. Effective poisons are closer to the target in gradient space
- C. Data points with isolated gradients lie in the nuisance space and slow down training
- D. Effective poisons are closer to the target in embedding space
- E. Only a fraction of the poisons can be optimized by bounded perturbations.
- ☒ F. The gradients of effective poisons are isolated

1:

$$1) \quad x_1 = [1, 1, 3], \quad x_1' = [2, 1, 3]$$

$$\|x_4 - x_1'\|_2 = \sqrt{6}$$

$$2) \quad x_2 = [1, 2, 2], \quad x_2' = [2, 2, 2]$$

$$\|x_4 - x_2'\|_2 = \sqrt{14}$$

$$3) x_3 = [2, 1, 0], x_3' = [3, 1, 0]$$

$$\|x_4 - x_3'\|_2 = \sqrt{26}$$

based on smallest ℓ_2 distance, choose x_1 .

To keep clean label attack, keep $y_1 = 1$

$$2: x_3 = [2, 1, 0] \quad x_4 (\text{target}) = [3, 0, 5]$$

went to use perturbation to decrease ℓ_2 distance:

$$x_3' = [2.5, 0.5, 0.5]$$

3: From lecture 8:

"Effective poison gradients are not similar to each other or clean data"

(F)